Algorithm: Milk Collection, Profit Calculation, and Vaccination Reminder System

1. Start
2. Set VaccinationCount = 0
3. Read N (number of animals)
4. For Count = 1 to N:
   a. Read MorningMilk (re-ask if negative)
   b. Read EveningMilk (re-ask if negative)
   c. TotalMilk = MorningMilk + EveningMilk
   d. AmountEarned = TotalMilk * RatePerLiter
   e. Display TotalMilk, AmountEarned
   f. Read FeedingExpense (re-ask if negative)
   g. Profit = AmountEarned - FeedingExpense
   h. Display Profit
   i. Read LastVaccinationDate
      If (next vaccination due within 30 days):
         Display "Vaccination Needed"
         VaccinationCount++
      Else:
         Display "Vaccination Not Required"
   j. End of loop
5. Display final report (totals, profit, VaccinationCount)
6. End

CODE

```
#include <stdio.h>
#include <time.h>
#define RATE_PER_LITER 45.0   /* Rate per liter of milk */

int main() {
    int n, count, vaccinationCount = 0;
    char choice;

    printf("Enter number of animals: ");
    scanf("%d", &n);

    /* Arrays to store final report data */
    float totalMilkArr[n], earnedArr[n], expenseArr[n], profitArr[n];
```

```
for (count = 1; count <= n; count++) {
    float morningMilk, eveningMilk, totalMilk, amountEarned, feedExpense, profit;
    int age;

    printf("\n--- Animal %d ---\n", count);

    /* ---------- Morning Milk ---------- */
    do {
        printf("Enter morning milk quantity (liters): ");
        scanf("%f", &morningMilk);

        if (morningMilk < 0) {
            printf("Invalid value. Skip this entry and set to 0? (y/n): ");
            scanf(" %c", &choice);
            if (choice == 'y' || choice == 'Y') {
                morningMilk = 0;
                break;
            }
        } else {
            break;
        }
    } while (1);

    /* ---------- Evening Milk ---------- */
    do {
        printf("Enter evening milk quantity (liters): ");
        scanf("%f", &eveningMilk);

        if (eveningMilk < 0) {
            printf("Invalid value. Skip this entry and set to 0? (y/n): ");
            scanf(" %c", &choice);
            if (choice == 'y' || choice == 'Y') {
                eveningMilk = 0;
                break;
            }
        } else {
            break;
        }
    } while (1);

    /* ---------- Total Milk & Earnings ---------- */
    totalMilk = morningMilk + eveningMilk;
    amountEarned = totalMilk * RATE_PER_LITER;
```

```
    printf("Total milk collected: %.2f liters\n", totalMilk);
    printf("Amount earned: %.2f\n", amountEarned);

    /* ---------- Feeding Expense ---------- */
    do {
        printf("Enter feeding expense: ");
        scanf("%f", &feedExpense);

        if (feedExpense < 0) {
            printf("Invalid value. Skip this entry and set to 0? (y/n): ");
            scanf(" %c", &choice);
            if (choice == 'y' || choice == 'Y') {
                feedExpense = 0;
                break;
            }
        } else {
            break;
        }
    } while (1);

    /* ---------- Profit ---------- */
    profit = amountEarned - feedExpense;
    printf("Profit: %.2f\n", profit);

    /* Store values for final report */
    totalMilkArr[count - 1] = totalMilk;
    earnedArr[count - 1]    = amountEarned;
    expenseArr[count - 1]   = feedExpense;
    profitArr[count - 1]    = profit;

    /* ---------- Vaccination Check ---------- */
    int lastVaccDay, lastVaccMonth, lastVaccYear;


printf("Enter last vaccination date (DD MM YYYY): ");
scanf("%d %d %d", &lastVaccDay, &lastVaccMonth, &lastVaccYear);

// Create structure for last vaccination date
struct tm lastVacc = {0};
lastVacc.tm_mday = lastVaccDay;
lastVacc.tm_mon = lastVaccMonth - 1;   // Months are 0-11
lastVacc.tm_year = lastVaccYear - 1900; // Years since 1900

// Convert to time_t
```

```
    time_t lastVaccTime = mktime(&lastVacc);

    // Get current date and time
    time_t currentTime = time(NULL);

    // Calculate difference in seconds and convert to days
    double diffSeconds = difftime(currentTime, lastVaccTime);
    int daysSinceLastVacc = (int)(diffSeconds / (60 * 60 * 24));

    printf("Days since last vaccination: %d\n", daysSinceLastVacc);

    // Assuming vaccination interval is 30 days
    if (daysSinceLastVacc >= 30) {
        printf("Vaccination Needed\n");
        vaccinationCount++;
    } else {
        printf("Vaccination Not Required\n");
    }

    printf("Vaccination Count: %d\n", vaccinationCount);

            }

    /* ---------- Final Report ---------- */
    printf("\n========== FINAL REPORT ==========\n");
    for (count = 1; count <= n; count++) {
        printf("\nAnimal %d:\n", count);
        printf("  Total Milk     : %.2f liters\n", totalMilkArr[count - 1]);
        printf("  Amount Earned  : %.2f\n", earnedArr[count - 1]);
        printf("  Feeding Expense: %.2f\n", expenseArr[count - 1]);
        printf("  Profit         : %.2f\n", profitArr[count - 1]);
    }
    printf("\nTotal animals requiring vaccination: %d\n", vaccinationCount);
    printf("===================================\n");

    return 0;
}
```

Output

```
Last login: Mon Jun 15 21:24:24 on ttys003
/Users/donalsaji/Desktop/Christ/Sem\ 1/C\ Progrmming/Lab/Lab1-3 ; exit;
donalsaji@Donals-MacBook-Pro ~ % /Users/donalsaji/Desktop/Christ/Sem\ 1/C\ Progrmming/Lab/Lab1-3 ; exit;
Enter number of animals: 4

--- Animal 1 ---
Enter morning milk quantity (liters): 7
Enter evening milk quantity (liters): 8
Total milk collected: 15.00 liters
Amount earned: 675.00
Enter feeding expense: 379
Profit: 296.00
Enter last vaccination date (DD MM YYYY): 12 05 2026
Days since last vaccination: 35
Vaccination Needed
Vaccination Count: 1

--- Animal 2 ---
Enter morning milk quantity (liters): 8
Enter evening milk quantity (liters): 3
Total milk collected: 11.00 liters
Amount earned: 495.00
Enter feeding expense: 345
Profit: 150.00
Enter last vaccination date (DD MM YYYY): 12 06 2026
Days since last vaccination: 4
Vaccination Not Required
Vaccination Count: 1

--- Animal 3 ---
Enter morning milk quantity (liters): 6
Enter evening milk quantity (liters): 9
Total milk collected: 15.00 liters
Amount earned: 675.00
Enter feeding expense: 324
Profit: 351.00
Enter last vaccination date (DD MM YYYY): 12 05 2026
Days since last vaccination: 35
Vaccination Needed
Vaccination Count: 2

--- Animal 4 ---
Enter morning milk quantity (liters): 4
Enter evening milk quantity (liters): 7
Total milk collected: 11.00 liters
Amount earned: 495.00
Enter feeding expense: 234
Profit: 261.00
Enter last vaccination date (DD MM YYYY):
e
Days since last vaccination: 35
Vaccination Needed
Vaccination Count: 3

========== FINAL REPORT ==========

Animal 1:
  Total Milk     : 15.00 liters
  Amount Earned  : 675.00
  Feeding Expense: 379.00
  Profit         : 296.00
```

```
========== FINAL REPORT ==========

Animal 1:
  Total Milk     : 15.00 liters
  Amount Earned  : 675.00
  Feeding Expense: 379.00
  Profit         : 296.00

Animal 2:
  Total Milk     : 11.00 liters
  Amount Earned  : 495.00
  Feeding Expense: 345.00
  Profit         : 150.00

Animal 3:
  Total Milk     : 15.00 liters
  Amount Earned  : 675.00
  Feeding Expense: 324.00
  Profit         : 351.00

Animal 4:
  Total Milk     : 11.00 liters
  Amount Earned  : 495.00
  Feeding Expense: 234.00
  Profit         : 261.00

Total animals requiring vaccination: 3
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Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```